Assignment -2

Q1) Clousre properties

Given

Complement:

1. Take the DFA M for L
2. Swap finial and non-final states.
3. The new DFA accepts $\Sigma^* - L$, so the reselt is regular

Kleene Start:

1. Take the NFA N for L,
2. Add a new start S' that is also finial.
3. Add ε : S' → old start.
4. Add ε ; each old finial state → old start.
5. The new NFA accepts L^*

Reversal

1. Take the NFA/DFA for L.
2. Reverse every transition arrow.
3. Make the old start the only finial states.
4. Add a new start with ε-transitions to all old finial states
5. The new NFA acccpts L^R.

Substitution ($\Sigma = (a) = R_a$)

② ε - NFA for (a+b)(ab)

use Thompson's construction : build the (a+b)*, build the (ab)* block, then join them with ε.

Block 1 : (a+b)*

- $q_0 \to \varepsilon \to q_1$, $q_0 - \varepsilon \to q_7$
- $q_1 \to \varepsilon \to q_2$, $q_1 \to \varepsilon \to q_3$
- $q_2 \to a \to q_4$, $q_3 \to b \to q_5$
- $q_4 \to \varepsilon \to q_6$, $q_5 - \varepsilon \to q_6$
- $q_6 - \varepsilon \to q_1$ (loop), $q_6 - \varepsilon \to q_7$.

Block 2 ; (ab)*

- $q_7 - \varepsilon \to q_8$, $q_7 \to \varepsilon \to q_{11}$
- $q_8 - a \to q_9$, $q_9 - b \to q_{10}$
- $q_{10} - \varepsilon \to q_8$ (loop), $q_{10} - \varepsilon \to q_{11}$

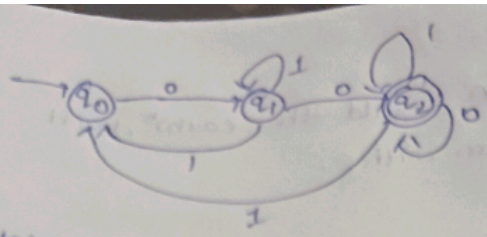
Start state : q_0 final state : q_{11}

③ (0+1)*00

The language is all binary string that end with 00.

$L = \{ w00 \mid w \in \{0,1\}^* \}$

L	Not in L
00	ε
100	0
1100	01
0000	001
10100	1001



State	0	1	Meaning
$\to q_0$	q_1	q_0	no trailing 0
q_1	q_2	q_0	one trailing 0
$*q_2$	q_2	q_0	two or more trailing 0s

- Start : q_0
- Final : q_2

1100: $q_0 \to 1 \to q_0 \to 1 \to q_0 \to 0 \to q_1 \to 0 \to q_2$ ✓

001 : $q_0 \to 0 \to q_1 \to 0 \to q_2 \to 1 \to q_0$ ✗.

(4)

Pumping lemma: $L = \{w : \#a = \#b\}$

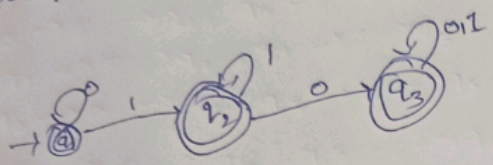
- Assume L is regular, with pumping length p.
- Take $s = a^p b^p$. The $|s| = 2p \geq p$ and $\#a = \#b$, so $s \in L$
- $s = xyz$ $|xy| \leq p$ $|y| \geq 1$.
- So $y = a^k$, $1 \leq k \leq p$.
- pump up with $i = 2$, $xy^2z = a^{p+k} b^p$.
- $\#a = p+k$, $\#b = p$ and $p+k \neq$ because $k > 1$
- So $xy^2z \in L$, which contradicts the pumping le
- $\therefore L$ is not regular.

s = aaa...a | bbb...b
x | y | z

|xy| ≤ p (all a's)

pump i=2 → a^(p+k) b^p x ∈ a* b*

⑤



R = Q + RP ⇒ R = QP*.

- $q_1 = \varepsilon + q_1 \cdot 0$
- $q_2 = q_1 \cdot 1 + q_2 \cdot 1$
- $q_3 = q_2 \cdot 0 + q_3 \cdot (0+1)$

$q_1 = \varepsilon + q_1 \cdot 0 \Rightarrow q_1 = 0^*$

$q_2 = 0^*1 + q_2 \cdot 1 = 0^*1 \cdot 1^* = 0^*1^+$

$q_3 = 0^*1^+0 + q_3(0+1) \Rightarrow q_3 = 0^*1^+0(0+1)^*$.

$RE = q_1 + q_2 + q_3 = 0^* + 0^*1^+ + 01^*0(0+1)^*$

$= 01(\varepsilon + 0(0+1)^*)$

$= (0+1)^*$

Only q_1 and q_2 final RE $= 0^* + 0^*1 = 0^*1^*$.